
L.S. COMPUTER TECHNOLOGH

SOFTWARE DEVELOPMENT & DIGITAL SOLUTIONS

PROJECT PROPOSAL

Maligathanna Archaeological Heritage Multi-Site Digital Portal & Content Management System (CMS)

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Date	September 2026
Version	1.0 – Initial Proposal
Submitted to	Maligathanna Archaeological Heritage Authority

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1. Executive Summary

L.S. COMPUTER TECHNOLOGH is pleased to present this formal project proposal for the development of the Maligathanna Archaeological Heritage Multi-Site Digital Portal and Content Management System. This proposal outlines the complete conversion of an existing static heritage website into a robust, database-driven web application — purpose-built to preserve, present, and promote Sri Lanka's rich archaeological legacy in a modern digital environment.

The proposed platform will empower authorised administrators and archaeological officers to manage all website content — across multiple heritage sites — without requiring any knowledge of source code or programming. It is designed with scalability, security, and cultural sensitivity at its core, supporting trilingual content in Sinhala, English, and Tamil.

This is not simply a website upgrade. It is the creation of a dedicated digital infrastructure for archaeological heritage management — one that can grow alongside the institution's needs, accommodate new sites over time, and serve both local communities and international researchers.

Item	Details
Project Name	Maligathanna Archaeological Heritage Multi-Site Digital Portal & CMS
Client	Maligathanna Archaeological Heritage Authority
Developer	Lakindu Jayathilaka – L.S. COMPUTER TECHNOLOGH
Technology Stack	PHP 8.2+, MySQL/MariaDB, JavaScript ES6+, Vanilla CSS, PWA
Languages Supported	Sinhala (සිංහල), English, Tamil (தமிழ்)
Heritage Sites Supported	Up to 20 Sites (Initial Package)
Development Investment	LKR 80,000 (All-inclusive)
First-Year Maintenance	Complimentary

2. Project Background

The Maligathanna Archaeological Heritage site currently maintains an online presence through a static website developed using HTML, CSS, JavaScript, and Progressive Web App (PWA) technology. While this existing platform effectively communicates the site's cultural significance, its static nature imposes significant operational limitations: every update to content, images, or information requires direct code modification by a technical developer.

As the site's profile grows and its informational needs become more complex — accommodating multiple heritage branches, trilingual audiences, and rich media archives — a purpose-built, dynamic content management system has become essential. The proposed redevelopment addresses these challenges comprehensively.

Current Limitations

- Content updates require direct file editing by a developer
- No administrative interface for non-technical staff
- Unable to scale to multiple heritage branch sites
- No multilingual content management workflow
- No structured media library or photo management
- No audit trail or access control for content contributors

Proposed Solution

The new platform will retain the cultural aesthetic and PWA capabilities of the existing website while introducing a complete backend CMS, relational database architecture, and a scalable multi-site framework. All existing content and design sensibilities will be preserved and enhanced during the transition.

3. Project Objectives

The primary objectives of this development project are as follows:

- Develop a fully dynamic, database-driven heritage management platform built on PHP 8.2+ and MySQL/MariaDB
- Create a secure, user-friendly CMS that allows non-technical administrators to manage all website content independently
- Implement a scalable multi-site architecture capable of supporting up to 20 distinct heritage sites under a single platform
- Deliver comprehensive trilingual support for Sinhala, English, and Tamil — with administrator-managed translations
- Build a structured media management system with WebP optimisation, thumbnail generation, and photographer attribution
- Establish role-based access control with audit logging to ensure content accountability
- Preserve and enhance the PWA functionality, offline capabilities, and responsive mobile design of the existing site
- Provide a long-term, maintainable digital platform that can grow with the institution's archival and outreach needs

4. Scope of Work

The development scope covers four primary components: the public-facing heritage website, the CMS administration panel, the multi-site management framework, and the technical infrastructure.

4.1 Public-Facing Heritage Website

Core Website Sections

- Professional homepage with cultural heritage design language
- Site introduction and heritage significance overview
- Detailed historical background and period narratives
- Archaeological findings and excavation information
- Architectural remains and structural documentation
- Interactive historical timeline with multilingual entries
- Filterable image gallery with categories
- Nearby heritage locations with distances and descriptions
- Google Maps integration and GPS navigation links
- Preservation status and conservation information
- Official credits and acknowledgements

Multilingual System

- Full trilingual support: Sinhala (සිංහල), English, Tamil (தமிழ்)
- Language switching from the public website interface
- Administrator-managed translations via the CMS
- Language-specific content for all text fields

Media Gallery

- Single and batch image upload with progress indicators
- Automatic WebP conversion and thumbnail generation
- Image categorisation and ordering controls
- Photo captions and photographer attribution fields
- Featured image designation per gallery section

Interactive & PWA Features

- Fully responsive design for all screen sizes
- Dark mode / light mode toggle with user preference memory
- Mobile-friendly navigation
- Text-to-Speech accessibility compatibility
- Progressive Web App (PWA) with installable interface
- Offline content caching via Service Worker

4.2 Administration Panel / CMS

Dashboard

- At-a-glance statistics: total sites, images, events, administrators
- Quick-access action buttons for common tasks
- Recent activity feed

Content Management

- Add, edit, publish, and archive content for each heritage site
- Manage Sinhala, English, and Tamil content in a unified interface
- Publish or hide specific sections independently

Media Management

- Upload and organise images with category tagging
- Edit captions, photographer information, and ordering
- Bulk delete and reassign image categories

Timeline Management

- Create and edit historical events with dates and descriptions
- Manage multilingual descriptions for each event

- Drag-and-drop or numeric display ordering

Nearby Attractions Management

- Add, edit, and remove nearby heritage locations
- Set distance, historical era, description, and Google Maps link
- Upload representative images for each location

User Management

- Super Admin: full platform access and user management
- Editor: content creation and media management
- Archaeology Officer: designated content review and publishing rights

Security Features

- Secure login with session management and timeout
- Password hashing (bcrypt) and secure credential storage
- PDO prepared statements — full SQL injection protection
- XSS (Cross-Site Scripting) output sanitisation
- CSRF (Cross-Site Request Forgery) token validation
- Role-Based Access Control (RBAC) across all admin functions
- Comprehensive audit logs with timestamp and user attribution

5. Multi-Site Architecture

One of the distinguishing capabilities of this platform is its multi-site design. Rather than developing separate, isolated websites for each archaeological or heritage location, the system uses a centralised database and shared codebase that can serve multiple heritage sites from a single domain and hosting environment.

This approach delivers significant practical advantages: lower long-term maintenance costs, a single point of updates for core functionality, and a unified administrative interface for managing all sites simultaneously. Each heritage site retains full content independence — its own name, description, images, timeline, GPS coordinates, and nearby attractions — while benefiting from shared infrastructure.

Capability	Details
Sites Supported (Initial)	Up to 20 Heritage Sites
Per-Site Data	Name, history, images, timeline, GPS, attractions, descriptions
Scalability	New sites added via CMS — no code changes required
Hosting Model	Single domain / hosting environment
Administrative Access	Centralised CMS managing all sites
Content Independence	Each site has fully independent content and media

6. Technical Architecture

The system will be built on a clean, modular PHP MVC-inspired architecture. This structure ensures long-term maintainability, clear separation of concerns, and straightforward onboarding for future developers.

Technology Stack

Layer	Technology
Server-side Language	PHP 8.2+
Database	MySQL / MariaDB (Relational)
Frontend	Vanilla CSS, JavaScript ES6+
Database Access	PDO with Prepared Statements
PWA	Service Worker, Web App Manifest
Image Processing	PHP GD / Imagick (WebP, Thumbnails)
Session Security	Secure PHP Sessions, CSRF Tokens

Application Structure

- /config — Database connection, environment constants
- /controllers — Request handling and business logic
- /models — Data access and database interaction
- /views — Modular PHP/HTML templates
- /helpers — Utility functions (sanitisation, formatting)
- /admin — CMS administration panel
- /assets — CSS, JavaScript, fonts, icons
- /uploads — Media file storage (organised by site)
- /api — JSON API endpoints for dynamic frontend calls

7. Implementation Phases

Phase	Title	Key Deliverables	Duration
1	Planning & Architecture	Requirements finalisation, database schema design, project structure setup	1 Week
2	Database & Backend Core	MySQL schema, PHP models, PDO integration, multi-site data layer	1.5 Weeks
3	CMS / Admin Panel	Admin dashboard, all CRUD modules, user management, security features	2 Weeks
4	Public Website Frontend	All public pages, gallery, timeline, multilingual interface, maps	1.5 Weeks

5	PWA & Performance	Service Worker, offline mode, WebP optimisation, mobile testing	0.5 Week
6	Content Migration & Testing	Content import from existing site, full QA, cross-browser testing	1 Week
7	Deployment & Handover	Live server deployment, admin training, documentation handover	0.5 Week
Total		End-to-end development and go-live	~8 Weeks

8. Deliverables

Upon completion of the agreed development scope and receipt of the final payment, the following deliverables will be provided to the client:

- Fully functional multi-site heritage web application deployed on the client's hosting environment
- Secure CMS administration panel with all agreed management modules
- Trilingual content management interface (Sinhala, English, Tamil)
- Media library with WebP optimisation and categorisation
- Configured Progressive Web App with offline Service Worker
- MySQL database with complete schema and initial content population
- Role-based user access system with audit logs
- Google Maps and GPS navigation integration
- Administrator user guide / documentation (PDF format)
- One-time administrator onboarding session
- Source code (upon final payment)

9. Hosting & Domain Information

The proposed platform will be hosted with LankaHost, a trusted Sri Lankan hosting provider well-suited for PHP and MySQL-based applications. Domain registration will be handled through register.domains.lk. Please note that all hosting and domain charges are third-party costs independent of the development fee.

Item	Provider	Estimated Cost
Web Hosting (Annual)	LankaHost	LKR 6,000 / year
Domain Registration	register.domains.lk	Varies by domain name
Hosting Renewal	LankaHost	Subject to provider rates
Domain Renewal	register.domains.lk	Subject to provider rates

Important Notes Regarding Hosting & Domain

- Hosting and domain charges are payable directly to the respective third-party providers and are not included in the development fee.
- Domain registration pricing varies based on the chosen domain name and extension (e.g., .lk, .com, .org).
- Renewal prices for both hosting and domain are subject to change at the discretion of the respective providers.
- A single hosting environment can be structured to serve multiple heritage branch websites under this platform.

10. Project Pricing

The following investment covers the complete design, development, system architecture, CMS implementation, copywriting support, content structuring, and reasonable scope adjustments throughout the development process.

Service Included	Detail
System Architecture & Database Design	Included
PHP Backend & CMS Development	Included
Public-Facing Heritage Website	Included
Multilingual Support (Sinhala, English, Tamil)	Included
Media Library & WebP Optimisation	Included
PWA & Service Worker Implementation	Included
Security Implementation (CSRF, XSS, RBAC)	Included
Copywriting Support & Content Structuring	Included
Up to 20 Heritage Sites (Initial Package)	Included
Deployment & Administrator Handover	Included
TOTAL DEVELOPMENT INVESTMENT	LKR 80,000

♦ *Hosting and domain registration costs are NOT included in this figure and are payable separately to the respective providers.*

11. Payment Plan

Development payments are structured across two clear milestones to ensure both parties are protected throughout the project lifecycle.

Stage	Milestone	Description	Amount
1	Development Advance (50%)	Payable before development commences. Confirms project initiation and resource allocation.	LKR 40,000
2	Final Payment (50%)	Payable upon completion of all agreed deliverables and client acceptance.	LKR 40,000

Total			LKR 80,000
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✦ Payment may be made via bank transfer or mobile payment. Details will be provided upon agreement.

12. Maintenance & Annual Renewal

To ensure the long-term health and performance of the platform, L.S. COMPUTER TECHNOLOGH provides ongoing maintenance support. The first year of standard maintenance is provided complimentary as part of the development package.

Period	Maintenance Coverage	Charge
Year 1 (Post-Launch)	Bug fixes, normal content changes, minor system adjustments	FREE
Year 2 Onwards (Annual)	Ongoing technical support, content updates, compatibility maintenance	LKR 20,000 / year

Maintenance Coverage Includes:

- Bug identification and resolution
- Normal content additions and updates (text, images, timeline events)
- Minor interface adjustments within existing design
- PHP and security compatibility checks
- Performance monitoring and basic optimisation

Exclusions & Additional Cost Scenarios:

- Annual hosting renewal charges — payable directly to LankaHost
- Annual domain renewal charges — payable directly to register.domains.lk
- Major new features or modules outside the original agreed scope
- Third-party plugin or API licensing fees (if applicable in future)

✦ Hosting and domain renewal prices are determined exclusively by the respective third-party providers and may change from year to year.

13. Development Team

Detail	Information
Company / Development Team	L.S. COMPUTER TECHNOLOGH
Main Developer	Lakindu Jayathilaka
Specialisation	Web Application Development, CMS Systems, Heritage Digital Platforms
Contact Number	+94 72 607 6637
Email Address	l.s.computertechnologh@gmail.com

L.S. COMPUTER TECHNOLOGH is a Sri Lanka-based software development service with hands-on experience in building dynamic web applications, learning management systems, and content-driven digital platforms. This project will be led personally by the main developer to ensure quality, continuity, and direct communication throughout the engagement.

14. Intellectual Property & Copyright Notice

© COPYRIGHT NOTICE

This project proposal document — including its concept, system architecture, design ideas, feature specifications, multi-site framework concept, content structure, and all associated intellectual ideas — is the original work of:

Lakindu Jayathilaka

L.S. COMPUTER TECHNOLOGH

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14.1 Ownership of Proposal & Ideas

This document constitutes an original creative and technical work prepared solely by Lakindu Jayathilaka of L.S. COMPUTER TECHNOLOGH. The proposal concept, system design ideas, multi-site heritage framework architecture, CMS feature structure, content workflows, and all associated intellectual property contained herein are the exclusive property of Lakindu Jayathilaka.

No part of this proposal — including its concept, structure, described features, or architectural ideas — may be reproduced, adapted, shared with third parties, or used as a basis for commissioning work with another developer or organisation without the express written consent of Lakindu Jayathilaka.

14.2 Protection of Concept & Innovation

The ideas, approach, and feature concepts described in this proposal represent original intellectual work developed specifically for this project. The following are protected under this copyright notice:

- The multi-site heritage digital platform concept and architecture design
- The proposed CMS structure, admin dashboard layout, and feature specifications
- The trilingual content management workflow and implementation approach
- The system design methodology, database architecture concept, and modular framework idea
- All written content, descriptions, scope definitions, and technical language within this document
- The proposal document itself as a complete original work

14.3 Permitted Use

This proposal is submitted exclusively for review by the named recipient — the Maligathanna Archaeological Heritage Authority — for the purpose of evaluating and engaging L.S. COMPUTER TECHNOLOGH for the described development work. Permitted uses are limited to:

- Internal review and evaluation by the named client organisation

- Discussion between the client and L.S. COMPUTER TECHNOLOGH regarding project scope and engagement
- Reference during an active, contracted development engagement with L.S. COMPUTER TECHNOLOGH

14.4 Restrictions

The following actions are expressly prohibited without prior written authorisation from Lakindu Jayathilaka:

- Sharing, forwarding, or distributing this document to any third-party developer, agency, or organisation
- Using the ideas, architecture, or feature concepts described herein to commission equivalent work from another party
- Reproducing any portion of this document in another proposal, tender, or planning document
- Copying or adapting the described system architecture for use in any other project without consent

14.5 Post-Development Intellectual Property

Upon completion of development and receipt of the final payment in full, ownership of the custom-developed source code, database schema, and client-specific content is transferred to the client as per the agreed scope. The following rights are retained by Lakindu Jayathilaka regardless of payment:

- Authorship credit and the right to reference this project in a professional portfolio
- Ownership of any reusable generic code components, base frameworks, and utility libraries developed as part of this project
- Ownership of the original proposal concept, idea, and document as a creative work
- The right to apply similar architectural approaches to future, unrelated projects

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15. Terms & Conditions

15.1 Proposal Validity

This proposal is valid for 30 days from the date of issue. Pricing and terms are subject to change after this period.

15.2 Payment Terms

- A non-refundable advance of LKR 40,000 (50%) is required before development commences.
- The remaining LKR 40,000 (50%) is due upon completion and client acceptance of the agreed deliverables.
- Development will not begin until the advance payment is received and confirmed.

15.3 Scope Changes

- Minor adjustments and reasonable content changes within the agreed scope are included.
- Significant new features or major scope additions outside the original agreement will require a separate written quotation.

15.4 Content & Assets

- The client is responsible for providing all original text content, historical information, and images in a timely manner.
- Delays in content delivery may affect the project timeline.

15.5 Hosting & Domain

- Hosting and domain registration are third-party services and are not covered under this development fee.
- The client is responsible for maintaining active hosting and domain subscriptions for the platform to remain operational.

15.6 Confidentiality

Both parties agree to keep all project-related information, technical specifications, and financial terms confidential during and after the engagement.

15.7 Limitation of Liability

L.S. COMPUTER TECHNOLOGH's liability is limited to the value of the development fee paid. The developer is not responsible for data loss, hosting outages, or domain-related issues caused by third-party providers.

16. Conclusion

The Maligathanna Archaeological Heritage Multi-Site Digital Portal represents an important step toward preserving and presenting Sri Lanka's cultural history in a way that is accessible, lasting, and administratively sustainable. This proposal puts forward a technically sound, professionally designed, and cost-effective path to achieving that goal.

L.S. COMPUTER TECHNOLOGH brings focused expertise in PHP-based web application development, CMS design, and heritage-conscious digital solutions. Every aspect of this project — from the trilingual content framework to the scalable multi-site architecture — has been designed with the long-term operational needs of archaeological and heritage institutions in mind.

We believe this platform will serve not only as a practical management tool for administrators, but as a dignified and enduring digital representation of Maligathanna's historical significance — available to researchers, tourists, and communities locally and internationally.

We welcome the opportunity to discuss this proposal further and are happy to address any questions regarding scope, timeline, or technical approach. We look forward to building something meaningful together.

Lakindu Jayathilaka

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This proposal was prepared exclusively for the Maligathanna Archaeological Heritage Authority by L.S. COMPUTER TECHNOLOGH. All pricing, terms, and scope details are confidential.